

Zero-shot object counting with LLM & VLM



Abstract

We introduce Zero-Shot Object Counting (ZSC), a challenging task that requires counting arbitrary objects without visual examples. Our core contribution is a novel method to **generate high-quality, image-specific counting exemplars on the fly**. We propose a framework where Vision-Language Models (VLMs) and Large Language Models (LLMs) collaborate to guide a generative process, synthesizing ideal object samples directly from a class name. These generated exemplars provide robust visual guidance, enabling accurate object counting. Experiments on the FSC-147 dataset validate that our approach of generating superior samples is highly effective.

Fresh Cut



Kiwi

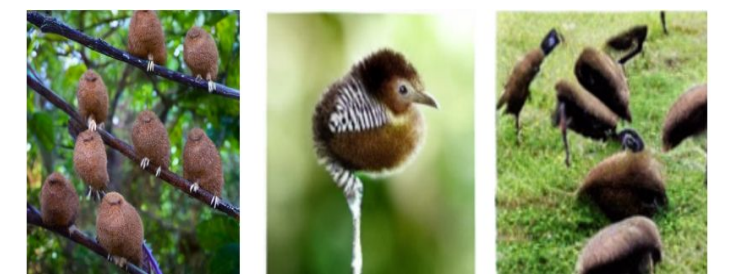


Fig 1. Wrong Exemplar generated by stable Diffusion

Proposed Method

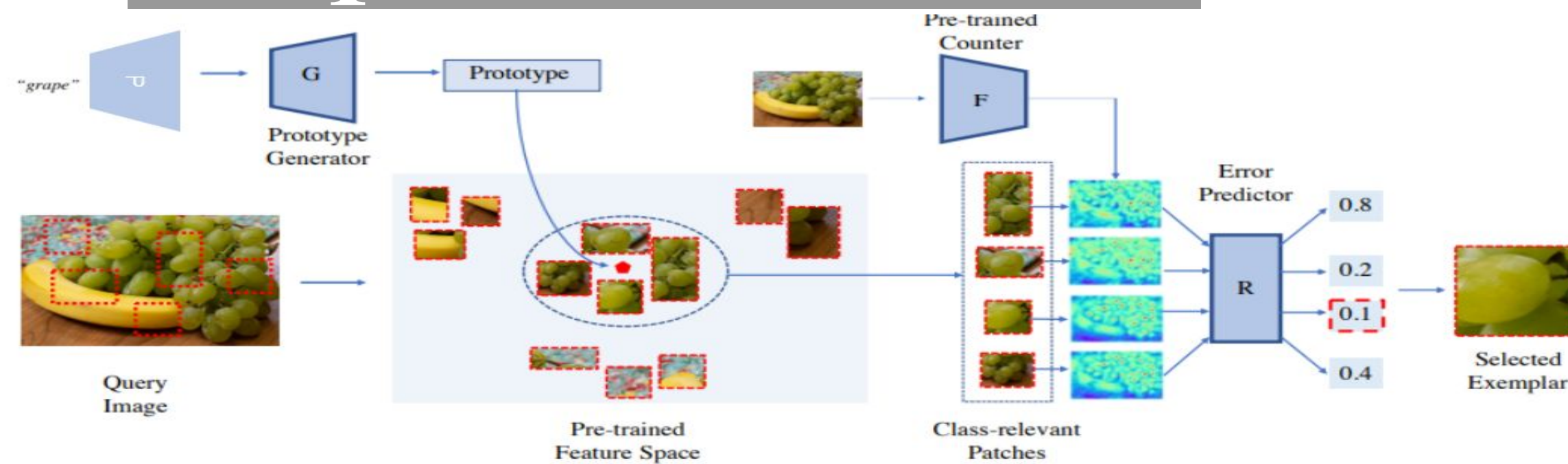


Fig 2. The whole pipeline in our work

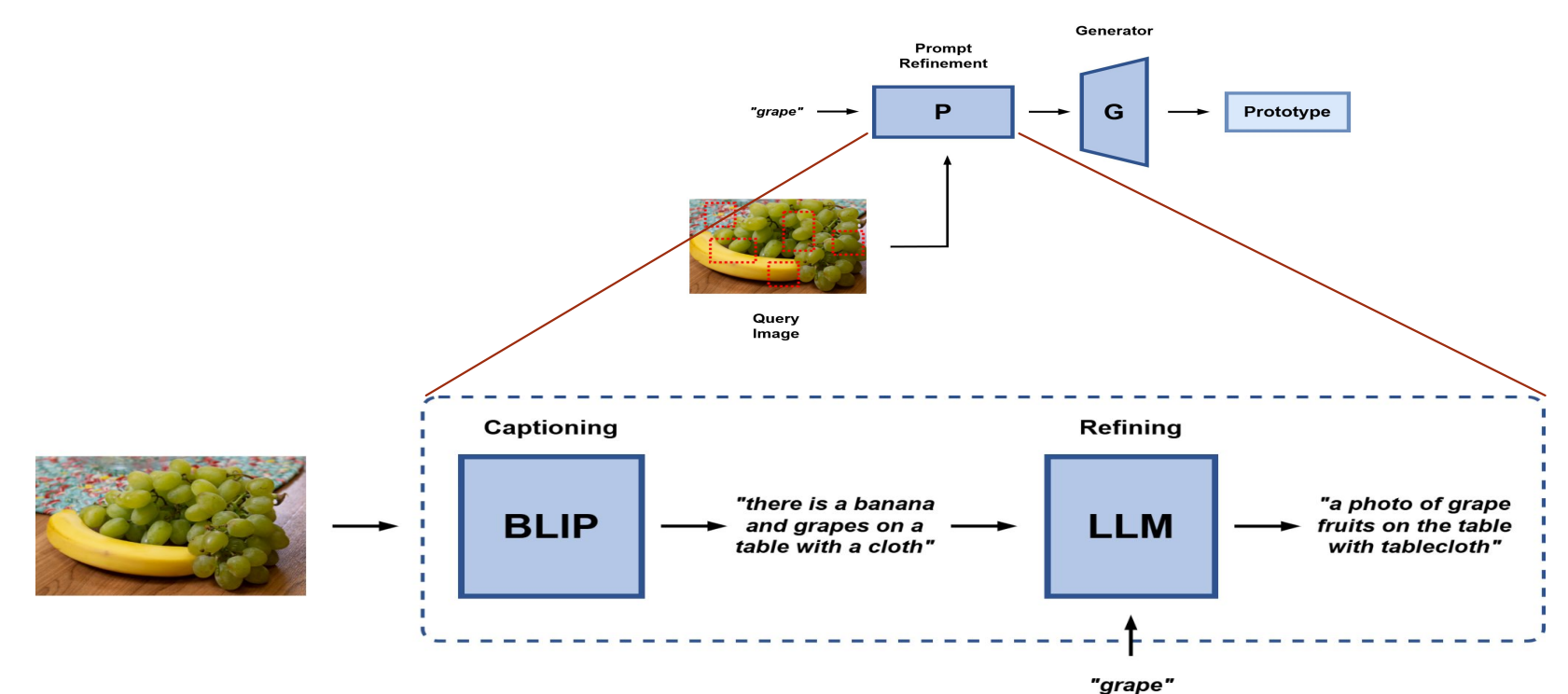


Fig 3. Architecture detail of P to guide generate model

We begin with a query image and a target class name (e.g., grape) as input. First, the BLIP model generates a caption for the query image, which is combined with the original class name and refined by LLM to produce a more accurate prompt (e.g., a close up of a bunch of green grapes hanging from a tree). This refined prompt is then fed into a prototype generator (G) to create a prototype image that the best represents the semantic meaning of the class in the given context. The query image is split into multiple patches, and each patch is compared with the prototype to select class-relevant patches. These selected patches are passed through a pre-trained counting model to predict the count of objects, while an error predictor evaluates each patch. Eventually, the system selects the most representative exemplar, improving the accuracy and generalization of the zero-shot counting task. By integrating vision-language models with semantic prompt refinement, our approach ensures that the generated prototypes and selected patches are both “contextually” and “semantically” relevant, leading to the best final predictions.

Experiment and Result

Our proposed architecture addresses both exemplar requirement challenges and word sense disambiguation issues, effectively transforming simple class labels (e.g., "kiwi") into semantically enriched target descriptions (e.g., "kiwi bird") through reinforced statement generation, which subsequently enables diffusion models to synthesize high-quality exemplar images. The experimental evaluation demonstrates significant performance improvements across 2 key metrics, Mean Absolute Error (MAE) and Root Mean Square Error (RMSE).

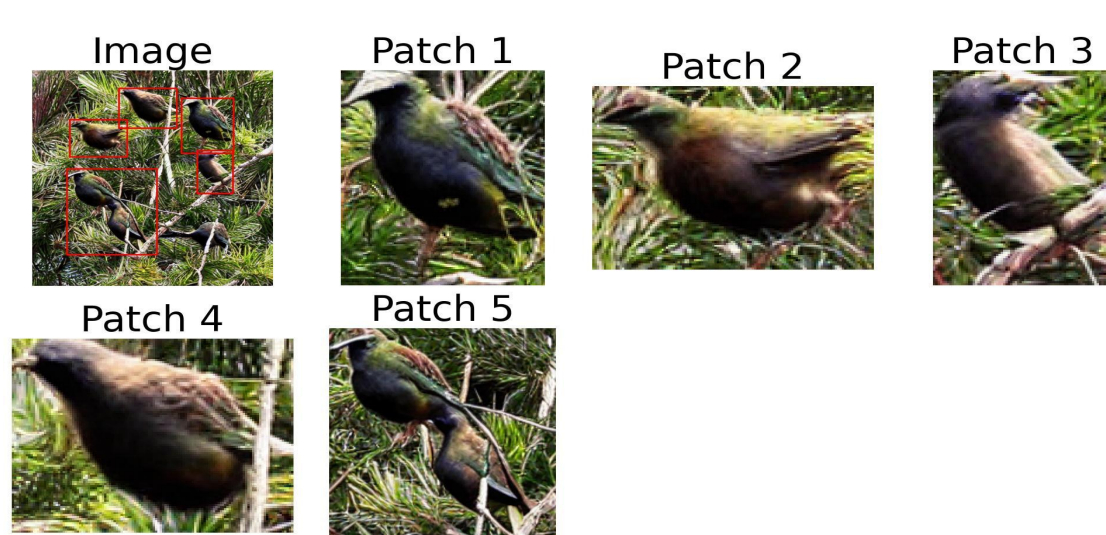


Fig 4. Generated kiwi bird exemplars using stable diffusion with semantically enhanced prompts.

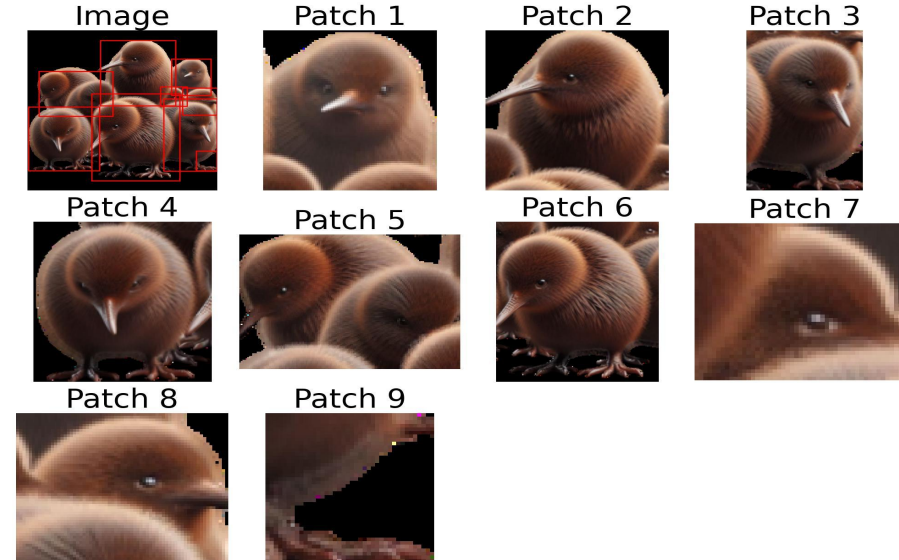


Fig 5. Quantitative results of kiwi bird counting accuracy on test images.

Image caption: a group of kiwi birds standing together on a black background
Original class: kiwis
Refined Prompt: a photo of Kiwi birds

Fig 6. Vision-Language Model (VLM) pipeline for converting simple class labels into semantically enhanced statements.

Conclusion

In this work, we introduced a novel prompt refinement strategy tailored for this task. Our approach refines textual prompts to better capture the intended semantics of target categories, leading to the **synthesis of more accurate and representative prototypes** through generative models. As a result, our proposed system achieves better performance compared to the original methods.

Limitation (Future work):

Efficiency via Model Distillation: To improve efficiency, use model distillation

Handle Complexity Use Positional Information: To handle complex scenes, we plan to incorporate positional information into prompts.

Evaluation Metrics

Metric	Baseline	Our Method	Improvement
MAE	21.52	19.31	-2.22 (-10.3%)
RMSE	113.28	105.77	-7.51 (-6.6%)

Table 1. Performance comparison between baseline and our proposed method. Our approach achieves 10.3% and 6.6% improvements in MAE and RMSE respectively.